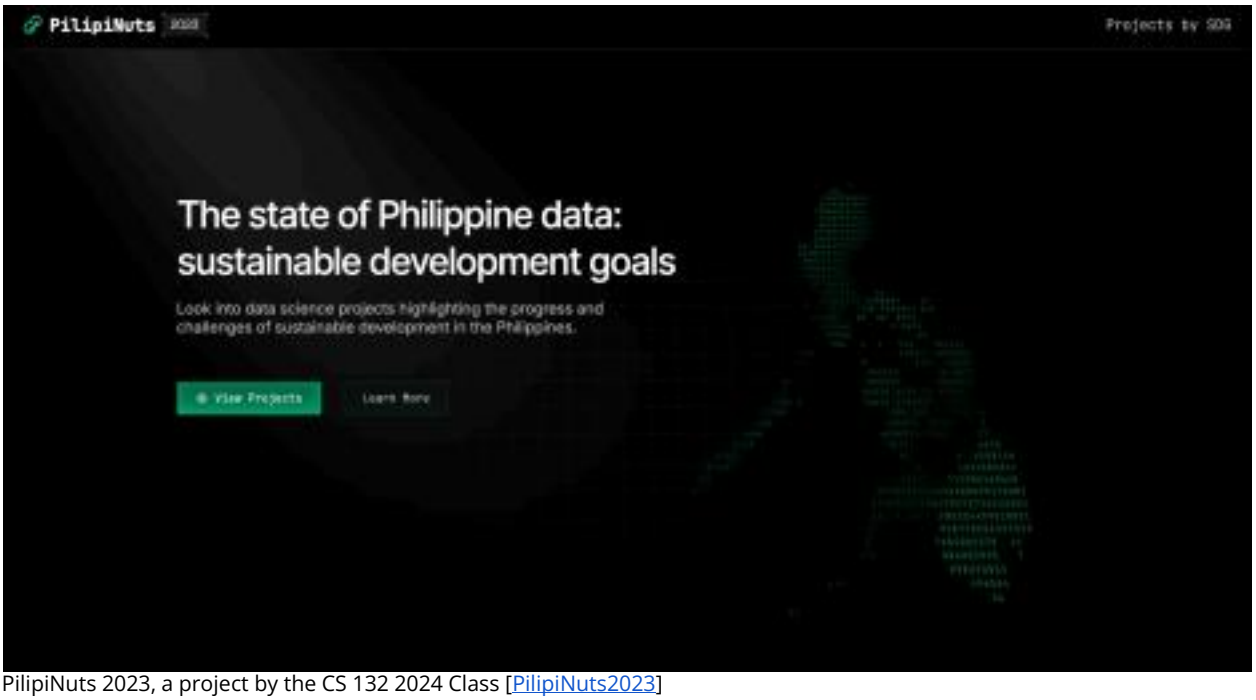
Pilipinas in a Nutshell (PilipiNuts) 2026

CS 132 Capstone Project Guidelines

The capstone project is the major requirement for CS 132 and will evaluate students' ability to apply learned concepts to real-world data. This semester, capstone projects will center around the theme "Pilipinas in a Nutshell (PiNut) 2026" where you'll study Philippine data in 2025/2026.

Objectives

1. Apply data science skills and techniques learned throughout the course to solve real-world problems or analyze real datasets.
2. Develop teamwork and collaboration skills through effective group work in planning, executing, and documenting a data science project.
3. Develop communication and presentation abilities by clearly explaining project findings, methods, and insights to both technical and non-technical audiences.



PilipiNuts 2023, a project by the CS 132 2024 Class [PilipiNuts2023]

Mechanics

Students will collaborate in groups to complete a data science project, covering tasks such as data collection, preprocessing, exploratory data analysis, and machine learning modeling. Each group will create a GitHub page to document and present their work and progress. The deadlines for project deliverables are listed here.

1. **Propose a research topic.**

   Your research topic must be unique and related to the UN Sustainable Development Goals (SDGs).

   When proposing a research topic, you will formulate at least 2 key research questions that you wish to answer through data science. For instance, under SDG 1—No Poverty, some interesting questions might be: *How has the poverty rate in the Philippines changed over the past decade? Which factors are linked to poverty rates? What is the impact of poverty on the mental well-being of Filipinos?* Note that your research question may relate to multiple SDGs, even if not specifically assigned to your class.

   In addition to the research topic, you will propose data sources from which you will gather the data needed for your research.

   > Caution: you are expected to produce original research. Copying published research without giving credit is not allowed and is considered academic dishonesty. It undermines the integrity of your work and can result in disciplinary action. Always give proper credit to the original authors to avoid plagiarism.



UN's 2030 Sustainable Development Agenda [Palsgaard]

SDG Assignments

Here are the SDG topics with sample areas of study:

1. **No Poverty**: Poverty Alleviation, Income Equality, Social Protection
2. **Zero Hunger**: Food Security, Agriculture Sustainability, Nutrition Improvement
3. **Good Health and Well-being**: Disease Prevention, Healthcare Access, Mental Health
4. **Quality Education:** Education Access, Skills Development, Lifelong Learning
5. **Gender Equality**: Women Empowerment, Gender-Based Violence Prevention, Equal Opportunities
6. **Clean Water and Sanitation**: Water Access, Sanitation Facilities, Water Quality
7. **Affordable and Clean Energy**: Renewable Energy, Energy Efficiency, Energy Access
8. **Decent Work and Economic Growth**: Job Creation, Fair Wages, Economic Development
9. **Industry, Innovation, and Infrastructure**: Infrastructure Development, Technological Innovation, Industrialization
10. **Reduced Inequality**: Social Inclusion, Income Distribution, Discrimination Prevention

11. **Sustainable Cities and Communities**: Urban Planning, Infrastructure Resilience, Environmental Sustainability
12. **Responsible Consumption and Production**: Waste Reduction, Sustainable Consumption, Resource Efficiency
13. **Climate Action:** Emission Reduction, Adaptation Strategies, Climate Resilience
14. **Life Below Water**: Marine Conservation, Sustainable Fisheries, Ocean Pollution
15. **Life on Land**: Biodiversity Conservation, Forest Protection, Land Restoration
16. **Peace, Justice, and Strong Institutions**: Conflict Resolution, Access to Justice, Governance Accountability

Project Proposal Format

Group Name:

Group Members:

Chosen SDG :

Project Title:

Research Questions

1.
2.
3.

Research Hypotheses

1.
2.
3.

Objectives

1.
2.
3.

Dataset & Resources

Grouping

You'll team up in groups of three or four with your classmates. Once you've formed a group and chosen your research topic, submit your proposal here. Indicate your group name, chosen Sustainable Development Goal (SDG), research topic, and data source on the spreadsheet.

2. **Collect and clean data.**
   See instructions below.

3. **Create and update your GitHub page.**
   See UVLE tutorial.

4. **Exploratory data analysis.**
   See instructions below.

5. **Train and evaluate your ML model.**
   Instructions to follow.

Grading

The project makes up 50% of your final grade. Your project score will be divided into three parts: data repository, exploratory data analysis, and machine learning modeling.

Data repository [10%]

Your initial task is to compile the data repository, which involves gathering a sufficient amount of data that is pertinent to your research question. Below are potential sources from which you can download the necessary data.

Sample data sources

The World Bank: website
Open Data PH (ODPH): website

Philippine Statistics Authority (PSA): website
ADB Data Library: website

Notes when collecting data

- Your dataset must contain data collected in 2025/2026. It can also include data from other years, but at least some data must be from 2025/2026.
- The data should be in raw form (i.e., not aggregated, such as averages). ● You must have at least 100 unique samples. Each observation counts as one sample; repeated observations over time are counted separately.
- Your data can be numeric, textual, visual (e.g., images), or a combination of these. ● Ensure you have the necessary permissions for data usage and publication if required. ● You are strongly advised to use existing datasets rather than collecting data yourself (e.g., through surveys).
- You may share the same dataset with other groups as long as you have different research questions.

Tip: Before proposing a research topic, check for available data. Given the project's tight schedule, you'll want to save time collecting data.

	3 pts	2 pts	1 pt
Data set size and quality	The dataset has at least 1000 observations without significant errors or inconsistencies.	The dataset has at least 500 observations, but there may be some issues with data quality or completeness, such as missing values.	The dataset has at least 100 observations but contains significant errors that impact its usefulness for analysis.

Data collection	The data collection method is implemented reliably, ensuring data integrity and minimizing biases or errors.	There may be some ambiguity or inconsistencies in implementation that could affect data quality.	Data collection methods are unclear or unreliable, leading to potential biases, inaccuracies, or uncertainties about the data's reliability and validity.
Relevance Documentation		The dataset is highly relevant to the research question, containing variables and observations directly applicable to the study objectives. Comprehensive documentation that details the data source, collection methods, and any preprocessing steps, ensuring transparency and facilitating replication of the study.	The dataset has some relevance to the research question, but there may be extraneous variables or data points that are less directly related to the study objectives. The documentation is adequate, but there may be gaps or inconsistencies in explaining the data source or collection procedures that could hinder understanding or reproducibility.
Deductions	**[-5 pts]** The dataset does not include data from 2024/2025 (or most updated data).		**[-1 pt]** Unclear/missing research objectives **[-1 pt]** Lacking background research **[-1 pt]** Unclear/unrelated research hypothesis **[-1 pt]** Unclear/missing proposed solution to the problem

Submission guidelines

- For your data collection document, you will need to submit the following:

    1. Description of your research questions/problems, proposed solutions, background, and hypotheses. These details should be included in your portfolio web page (see example). Note that you will only be graded for the contents, not the design yet.

    2. Description of data set, data collection process, sampling method (if applicable), data set size, preprocessing (if applicable), and other relevant information. These descriptions can be detailed on your portfolio web page, or on a Google doc if your web page is still under development. In such a case, add the Google Doc link to the web page.

    3. Lastly, a Google sheet containing all of your data (use separate sheets if your data comes from different sources). Make sure to add the Google sheet link to the web page.

- Once you are ready to submit your data collection document, create a PDF version of your portfolio webpage. You can simply use the print function on your web browser (the PDF version will only serve as a snapshot, so it's acceptable if the printed PDF version does not exactly look like your actual portfolio).

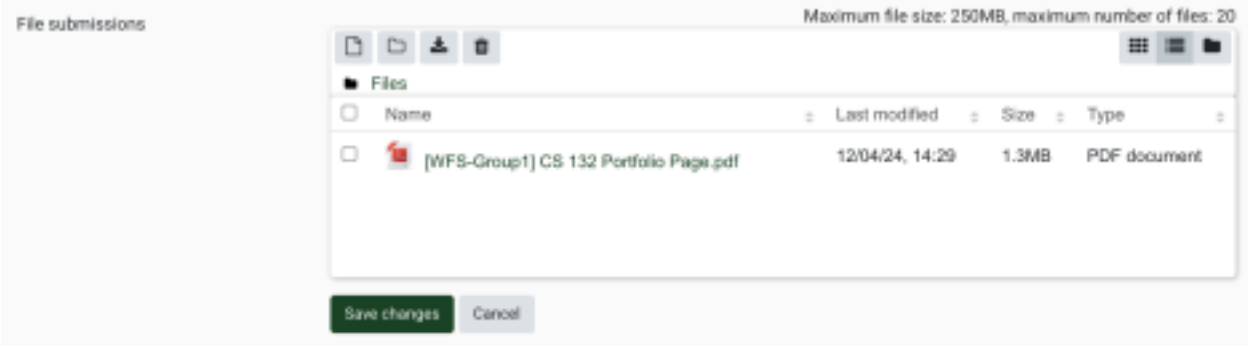
- Head over to UVLE, and submit the PDF to the ***Project Data Repository*** module. Each group member must make a submission.



- In the ***Project Data Repository*** submissions page, add the link to your web portfolio (e.g., Github Pages).



- Finally, upload the PDF version of your webpage with the filename:
  [Section-Group#] CS 132 Portfolio Page.pdf



Please refrain from editing your webpage after submission and before checking.

More notes

- You can share the same dataset with other groups as long as your research questions differ. However, for subsequent project components, you will be graded based on the originality of your analysis, so it's advisable not to exchange findings with other groups prior to submission.

- 2025 Data. If 2025 data is unavailable for your topic, you may use 2024 data, as long as you can show that it's the latest/most updated data.

Exploratory data analysis [20%]

Your next task is to visualize your data (in relation to your research questions) and test whether your hypothesis is correct.

Data Visualization

You will need to produce at least 3 graphs/plots:

- 1 for each of your research questions (total of 2)
- 1 "nutshell" plot

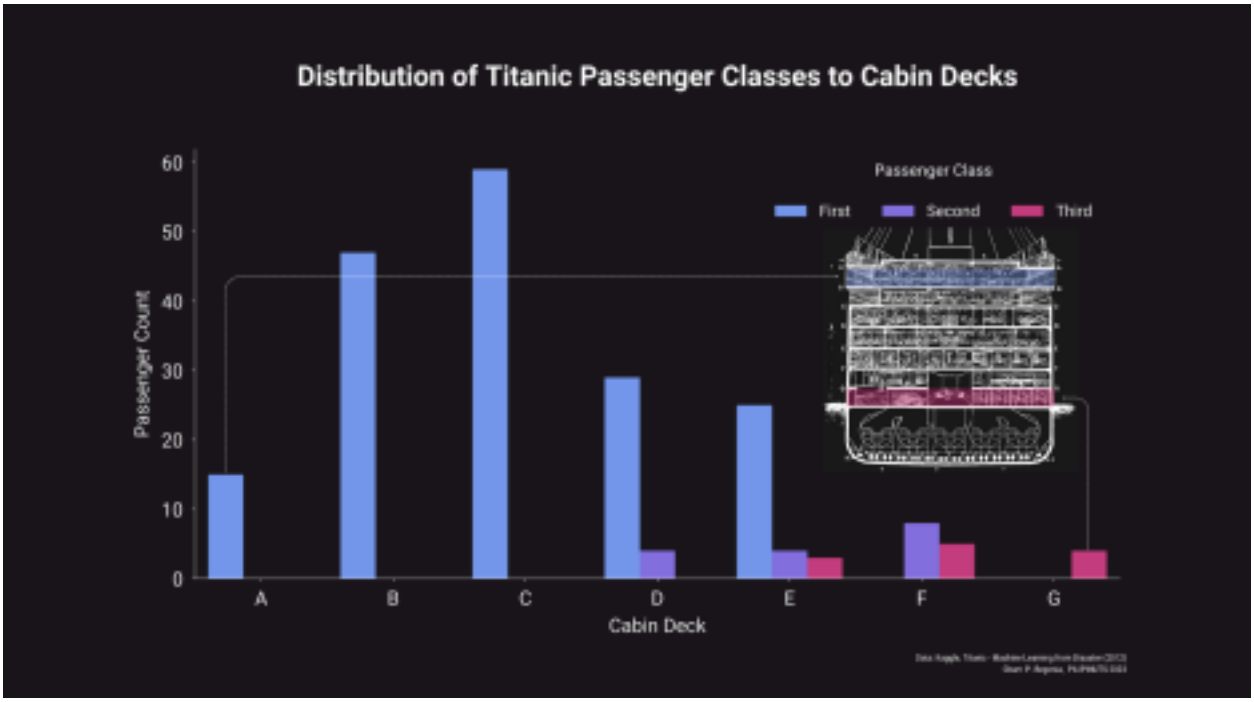
What is a nutshell plot?

A nutshell plot is a visual summary that highlights the key findings of your data science project. It answers your research questions clearly and includes essential insights to help readers understand your work. A good nutshell plot should be eye-catching, engaging, and accurate; it should tell the complete story.

For example, below is a nutshell plot for the research topic "*Analyzing the Impact of*

Social Class on Titanic Passenger Survival Rates".

[Headline] Sinking Inequality: A Titanic Tale of Class and Survival



Social class played a critical role in determining passengers' chances of survival during the Titanic disaster. The nutshell plot illustrates how passenger classes were distributed across cabin decks. Higher-class passengers were assigned to upper decks, which offered relatively safer conditions during the Titanic's sinking. In contrast, lower-class passengers were located on the lower decks, closer to the ship's bottom, where survival chances were significantly lower.

When creating your nutshell plots, you are required to follow the specs as detailed here:

- PilipiNuts 2024 Guidebook (Colab): link
- PilipiNuts 2024 Design Book (Canva): link

We will compile all your nutshell plots in the PilipiNuts 2024 portfolio. Here's the PilipiNuts 2023 portfolio for reference.

Hypothesis testing

You will perform statistical testing on any/all of your research hypotheses. Here's a guide on how to refine your hypotheses and select an appropriate test.

When choosing the hypothesis test:

1. Check whether your data is normally distributed
2. Follow the flow chart in the guide
    a. If your data is normally distributed, use the suggested parametric test
    b. If it's not, use a non-parametric version of the suggested test

	5 pts	3 pts	1 pt
Visualization (nutshell)	The nutshell graph effectively showcases the impact of the project, demonstrating clear visual communication and compelling storytelling, leaving a strong message	The nutshell graph presents several data points about the projects, conveying a decent level of data communication, analysis, and storytelling	The portfolio graph fails to effectively convey the project's significance, lacking clarity, coherence, or impact in presenting the data
Visualization (RQ 1)	answers the research question, incorporates multiple plots seamlessly without overcrowding, easy to understand	The visualization lacks visual appeal, interactivity, complexity, clarity, and/or coherence	
Visualization (RQ 2) The visualization is appealing and interactive, clearly	The visualization is appealing and interactive, clearly answers the research question, incorporates multiple plots seamlessly without	The visualization lacks visual appeal, interactivity, complexity, clarity, and/or coherence	
		overcrowding, easy to understand	
Hypothesis testing	The hypothesis testing is thorough, demonstrating a clear understanding of the hypothesis and an appropriate selection of statistical tests, leading to insightful conclusions	The hypothesis testing is satisfactory, showing a basic understanding of the hypothesis, using suitable statistical tests, and offering some interpretation of the	The hypothesis testing is inadequate, displaying a limited understanding of the hypothesis, employing incorrect or inappropriate statistical tests, and resulting in

		results	superficial or inconclusive findings
Data cleaning			Thorough data verification, cleaning, and preprocessing were done to ensure the data's cleanliness.
Documentation		The portfolio includes a functional public web page featuring a clear and consistent design, along with well-documented processes that cover the project's introduction, data collection, and exploratory data analysis comprehensively.	The portfolio lacks a functional public web page, clear design consistency, and/or details about the project's introduction, data collection, and exploratory data analysis.
Deductions			**[-1 pt]** Missing project title **[-1 pt]** Missing project team details (About us)

Submission guideline

- Prepare the following deliverables:
    1. Nutshell plot (PDF, high res, 1920x1080 px, landscape)
    2. 2 project plots (at least 1 per research question)

3. Hypothesis testing results (for any of your hypotheses)
4. Portfolio snapshot (PDF)

- Update your online portfolio with Items 2 and 3 (1 is optional). Make sure to have a project title at the start, and an about us section at the end of your portfolio.
- Similar to the previous submission, upload the PDF version of your webpage with the filename:
  [Section-Group#] CS 132 Portfolio Page EDA.pdf
  E.g.: *[WFS-Group1] CS 132 Portfolio Page EDA.pdf*

- Upload your portfolio (PDF) to the ***Exploratory Data Analysis*** submissions page in UVLE once the module opens.

- Finally, upload your nutshell plots to your own repository. Share the link to the **Project Master Sheet (link will be provided later).** Additionally, provide a **short headline** (max 10 words) and a **caption** (max 150 words) explaining your nutshell plot.

  Check the **AI Index: State of AI in 13 Charts** for examples on writing headlines and captions.

Notes:

- The PilipiNuts 2025 design requirements (typesetting, colors, etc.) **only apply to the Nutshell plot**. You are free to use your own graph designs for your group portfolios.

Modeling and interpretation [20%]

For your final tasks, your objective is to complete the remainder of your portfolio by: ● Writing a cohesive data analysis story from introduction to conclusion ● Providing an unbiased interpretation of your data science (EDA/ML) results ● Providing a recommendation/call to action based on your answers to your research questions

Machine learning (Optional)

You will develop a machine learning model to classify/predict/forecast/cluster significant

factors related to your research problem. Your goal is to construct an ML model that empowers Filipinos to make informed decisions by using current and historical data from the Philippines.

Although ML modeling is graded, **it is optional**, so you may opt not to submit/complete this part of the project.

Nutshell plot

You may resubmit your nutshell plots based on the comments on your EDA submission. If there are no comments, you will receive the full score by default.

Discussion and conclusion

In the latter section of your portfolio, you'll interpret your EDA and ML (if applicable) results, highlighting their relevance to your research problems and their potential socio-economic impacts on the Philippines. Furthermore, you'll address any limitations in your study concerning data availability and analysis techniques, and provide recommendations for future extensions of your research.

In your conclusion, you'll succinctly summarize your discoveries, explain the importance of your study, and provide actionable recommendations for Filipinos. Emphasize the significance of your research and illustrate how it can help Filipinos make informed decisions.

Web portfolio

Finally, your completed portfolio will be evaluated on the overall design and narrative. Ensure that every element is seamlessly integrated, well-designed, and consistent. Your goal is to weave a captivating narrative about the Philippines and inspire your readers to take action. Remember, data do not lie. What essential truth do you wish to impart to Filipinos?

	5 pts	3 pts	1 pt
ML modeling Nutshell plots (rework) Discussion	The choice of model is highly appropriate and well-justified based on the data and problem. Model assumptions and limitations are discussed. Hyperparameters are carefully tuned, and model performance is evaluated using relevant metrics. Insights are clearly tied to the problem statement, and practical implications are thoughtfully explored.	The selected model fits the problem reasonably well, with some justification. There may be limited exploration of alternatives or minimal hyperparameter tuning. Model performance is evaluated using standard metrics. Insights are provided but may lack depth or a strong connection to the problem. The (re)submitted nutshell plot is clear and well-designed, and adheres to the design guidelines. The discussion is thorough and provides a deep understanding of the EDA/ML results. It clearly explains the significance of the findings, connects them to the initial problem statement, and discusses implications. The	The model selection appears arbitrary or poorly suited to the problem. Little or no justification is provided. Performance evaluation is minimal or missing. Insights and implications are unclear or absent. The (re)submitted nutshell plot is unclear, poorly designed, or does not adhere to the design guidelines. The discussion is superficial and does not provide a clear understanding of the EDA/ML results. It may fail to connect the findings to the problem statement, overlook the significance and implications, and ignore limitations and potential improvements.

		limitations of the analysis and potential next steps or improvements are also addressed.	
Conclusion		The conclusion effectively summarizes the key findings and insights from the project. It clearly articulates the impact and significance of the results, tying them back to the original problem	The conclusion summarizes the main findings and insights, but may lack depth or clarity in articulating their impact and significance. It generally ties back to the original problem

		statement and objectives. The conclusion also suggests practical applications, future work, or potential improvements in a thoughtful manner.	statement and objectives but may not suggest practical applications or future work as effectively.
Web portfolio design	The portfolio design is highly consistent, with a cohesive look and feel throughout. Navigation is intuitive, making it easy for users to find and understand content. The overall visual impact is strong, with a professional and visually appealing design that excites and engages the viewer. Attention to detail in	The portfolio design is generally consistent, but may have minor discrepancies in style or layout. The sections could be improved for better ease of use. The visual impact is adequate, with a clean and functional design, though it may lack elements that truly engage or excite the	The portfolio design lacks consistency, with noticeable discrepancies in style or layout. The sections are confusing or cumbersome, making it difficult for users to find and understand content. The overall visual impact is weak, with a design that appears unprofessional or

	layout, typography, and color scheme enhances the user experience.	viewer. Basic attention to detail is present in layout, typography, and color scheme.	visually unappealing. Limited attention to detail in layout, typography, and color scheme detracts from the user experience.
Web portfolio story	The portfolio presents a compelling and coherent story that effectively ties together the whole project and its sections. The narrative flows logically, engaging the viewer from start to finish. Each project and section builds on the previous one, contributing to a unified theme or message. The portfolio includes a clear and persuasive call to action.	The portfolio tells a generally coherent story with minor gaps or inconsistencies. It engages the viewer but may not fully captivate throughout. Most sections contribute to the overall theme, though some connections may be weaker. The story includes a call to action, though it may not be as clear or persuasive.	The portfolio lacks a clear or coherent story, with a disjointed narrative that fails to engage the viewer. Sections appear disconnected, and the narrative does not effectively highlight the important parts of the project. There is no clear call to action, leaving the viewer uncertain about the next steps.

Note: It is possible to earn more than 20 points. Any extra points you receive will be added to your overall project grade as bonus credit.

Submission guideline

- Follow the submission instructions for Exploratory Data Analysis. Use the ***Modeling and Interpretation*** module in UVLE.

Notes:

- To resubmit nutshell plots, use the ***Project Overview (resubmission)*** sheet in the project master sheet.
- Nutshell plots with a perfect score (5) in EDA may still be resubmitted for minor revisions. Check the comments for more details.